

Multi-Class Expert Routing for Zero-Shot Chest CT Analysis via Vision–Language Models

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Abstract. Manual annotation of 3D chest CT is labour-intensive, limiting the scalability of deep-learning diagnostic systems. We present a parameter-efficient framework that fine-tunes VILA-M3 on CT-RATE via LoRA and routes volumetric localisation queries to VISTA3D. Two contributions are introduced: (i) *multi-class expert routing* — a single adapter trained on 39,000 class-conditioned samples to emit query-specific routing tokens across five VISTA3D classes (lung tumor, heart, liver, aorta, lung); and (ii) *class-constrained inference* — a lightweight inference-time rewrite that closes the gap between routing format precision (100%) and class precision without retraining. A *format-vs-class precision* decomposition separates structural and semantic routing components, enabling arithmetic verification of the segmentation gap. On zero-shot pulmonary lesion localisation in LIDC-IDRI, the framework lifts mean Dice from 0.000 to 0.273, matching the per-scan expert ceiling. The liver routing class achieves 100% class precision without constraint, establishing proof of concept for query-conditioned multi-class expert delegation.

1 Introduction

Chest CT is a primary imaging modality for diagnosing thoracic diseases including pulmonary nodules, pneumonia, pleural effusion, and cardiovascular conditions. Despite advances in deep learning, high-performing models in this domain still depend on large datasets annotated with dense bounding boxes or pixel-level segmentation masks—an expensive process requiring considerable radiological expertise and time [1].

Vision-Language Models (VLMs) offer an alternative by learning visual representations from the natural pairing of medical images and clinical free-text reports, bypassing the need for explicit spatial labels [2]. Early medical VLMs such as MedCLIP [3] and BioViL-T [4] have demonstrated strong performance on chest X-ray classification and report grounding. However, extending these approaches to 3D CT volumes introduces fundamental challenges: processing full volumetric sequences of tokens is computationally prohibitive, and most VLM architectures lack mechanisms for precise 3D spatial reasoning [10].

VILA-M3 [10] addresses this problem by introducing an expert routing architecture within a VLM pipeline. Rather than processing raw 3D voxels, the VLM acts as an orchestrator that delegates spatial tasks to specialist expert models via the MONAI VLM Agent Framework [9]. This enables high-fidelity 3D analysis without native volumetric tokenisation. Despite this architectural advantage, VILA-M3 is pre-trained on general medical image-text data and has not been specifically adapted for thoracic CT interpretation.

This paper proposes a targeted instruction-tuning pipeline that fine-tunes VILA-M3 on CT-RATE [16], a large-scale dataset of paired chest CT volumes and radiology reports. Adaptation is

achieved efficiently via LoRA [15], requiring only a fraction of the parameters of full fine-tuning. The framework’s routing mechanism directs spatial localisation queries to VISTA3D [8], a versatile 3D CT segmentation expert that exposes a closed set of 127 anatomical and pathological classes, including a **lung tumor** class used here as the expert-side proxy for lung-lesion localisation. Zero-shot transfer is then evaluated on LIDC-IDRI [17], an entirely unseen benchmark of expert-annotated pulmonary nodules, alongside a case retrieval evaluation on the CT-RATE hold-out split.

Contributions. (1) **Multi-class expert routing:** a single LoRA adapter trained on 39,000 class-conditioned samples that emits query-specific routing tokens across five VISTA3D classes (**lung tumor, heart, liver, aorta, lung**), generalisable to any supported class at inference time without retraining. (2) **Class-constrained inference:** a lightweight mechanism that converts saturated format precision (100%) into effective class precision (100%) without modifying model weights, accompanied by a *format-vs-class precision* decomposition for analytical verification. (3) **First zero-shot VLM-routed lung lesion localisation on LIDC-IDRI:** mean Dice 0.273 on 115 expert-annotated scans without any pixel-level supervision. The framework is expert-agnostic.

2 Related Work

Contrastive VLMs such as CLIP [2], MedCLIP [3], and BioViL-T [4] learn from image–text pairs and achieve strong zero-shot classification on chest X-rays. Larger multimodal models including Flamingo [13], GPT-4V [5], and Med-Gemini [6] extend clinical reasoning but remain limited to 2D inputs. For volumetric CT, nnU-Net [7] and LUNA16-benchmarked [23] CNNs set strong supervised detection baselines; VISTA3D [8] extends this with a promptable 127-class 3D segmentor. VILA-M3 [10] builds on the VILA visual language architecture [11], LLaVA [18], and the transformer-based [12] LLaMA backbone [14] to introduce expert routing within the MONAI VLM Agent Framework [9]. Our work fine-tunes VILA-M3 on CT-RATE via LoRA [15] for thoracic-specific routing and performs the first zero-shot VLM segmentation evaluation on LIDC-IDRI.

3 Proposed Methodology

3.1 Dataset Configuration

The framework employs a dual-dataset strategy that strictly separates adaptation from evaluation:

- **CT-RATE** [16]: A large-scale dataset of over 50,000 reconstructed 3D chest CT volumes paired with structured radiology reports. CT-RATE serves as the training corpus for instruction-tuning. We operate on a randomly sampled subset of 9,000 volumes; a 1,000-volume held-out split is reserved exclusively for case retrieval evaluation and never seen during adaptation.
- **LIDC-IDRI** [17]: A reference database of 1,018 CT scans with expert-consensus nodule annotations from four independent radiologists. A subset of 115 scans with ≥ 3 -radiologist consensus nodules is used for evaluation; this dataset is withheld entirely from training and used only for zero-shot detection benchmarking.

3.2 Model Architecture and Dynamic Routing

The framework positions VILA-M3 as an intelligent orchestrator between the clinical query and specialist processing pipelines. Figure 1 provides an overview of the system.

The routing pipeline proceeds in three stages:

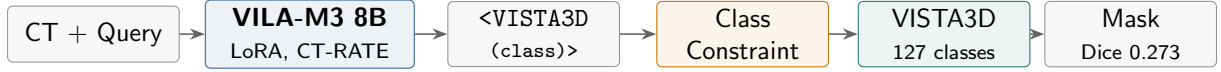


Figure 1: Framework overview. VILA-M3 (LoRA-adapted on CT-RATE) emits a structured routing token; class-constrained inference rewrites it to the task-relevant VISTA3D class, closing the format–class precision gap.

1. **Global Context Extraction.** VILA-M3 receives the user query (e.g., “*Identify and localise pulmonary abnormalities in this chest CT scan*”) together with a compact volumetric representation of the CT input, constructed by sampling key axial slices at uniform anatomical intervals.
2. **Task Classification and Routing.** The model classifies the query intent and emits a structured routing token in its output. Segmentation queries produce a token of the form `<VISTA3D(lung tumor)>`, which the MONAI VLM Agent Framework intercepts and dispatches to VISTA3D. Retrieval queries are handled internally through the VLM’s embedding space.
3. **Expert Execution and Response Synthesis.** VISTA3D processes the full-resolution CT volume with the `lung tumor` class prompt and returns a voxel-level segmentation mask. We note that this class is the closest lung-pathology label in VISTA3D’s 127-class vocabulary; it is trained on lung mass annotations rather than small nodule contours specifically, and is used here as a proxy expert for lung-lesion localisation (see Section 4.3 for the resulting per-scan ceiling and the proposed expert-substitution direction). The mask is evaluated directly against LIDC-IDRI ground truth for detection metrics. For retrieval queries, the VLM embedding is compared against the CT-RATE corpus index using cosine similarity.

This separation of concerns preserves the VLM’s semantic reasoning capacity while delegating computationally expensive volumetric processing to a purpose-built expert, avoiding the memory infeasibility of native 3D token processing.

3.3 Instruction Fine-Tuning Strategy

We apply Low-Rank Adaptation (LoRA) [15], a parameter-efficient fine-tuning strategy that builds on the adapter framework of Houlsby et al. [21] while reducing parameter overhead through low-rank matrix decomposition, to the VILA-M3 8B checkpoint (MONAI/Llama3-VILA-M3-8B). The query and value projection matrices of the language-model attention layers are targeted (the vision tower and multimodal projector remain frozen). The LoRA rank is set to $r = 16$ with a scaling factor $\alpha = 32$. Instruction data is constructed following the supervised fine-tuning protocol of Ouyang et al. [20]. A single adapter is trained *jointly* with two complementary objectives so that one fine-tuned model serves both downstream tasks:

- **Detection-routing objective (causal-LM loss).** Two training configurations are studied. In the *single-class* configuration, the adapter is trained on 9,000 CT-RATE pairs targeting the `lung tumor` routing class exclusively. In the *multi-class* configuration, approximately 39,000 instruction samples are derived from the same CT-RATE volumes by applying five class-specific query templates — `lung tumor`, `heart`, `liver`, `aorta`, and `lung` — together with descriptive negative samples that do not require routing. Each sample’s target response emits the appropriate class-conditioned routing token (e.g., `<VISTA3D(liver)>`, `<VISTA3D(heart)>`) when the paired report supports that interpretation, teaching the

VLM to bind the routing class to the *query intent* rather than to the image alone. In both configurations, the loss is the standard next-token cross-entropy on the response tokens, masking the prompt prefix and image-placeholder positions.

- **Retrieval objective (symmetric contrastive InfoNCE [19]).** For each volume in the batch we compute (i) an *image-side* embedding from the key slices paired with a neutral prompt and (ii) a *text-side* embedding from the radiology report, both taken as the last-token hidden state of the LLaMA body. Embeddings are ℓ_2 -normalised and compared via a CLIP-style symmetric InfoNCE loss with in-batch negatives, pulling each volume’s image embedding towards its own report and away from the other volumes in the batch. This is what enables embedding-space retrieval; no auxiliary text generation is used for retrieval.

Training uses a single NVIDIA A100 80 GB GPU (HPC AI-Center, Universitas Brawijaya) via the PyTorch cu118 stack. The AdamW optimiser runs at 1×10^{-4} with cosine annealing (3% warm-up), weight decay 0.01, effective detection batch size 4 (gradient accumulation), and contrastive in-batch size 8 at temperature $\tau = 0.07$. CT-RATE volumes are processed in a streaming fashion, with each volume discarded after slice extraction, enabling training on the full corpus without cumulative storage requirements.

4 Experimental Setup and Evaluation

4.1 Baselines and Metrics

Two baselines are used: (i) direct VISTA3D with `lung tumor` prompt (no VLM), establishing the per-scan ceiling; and (ii) pre-fine-tuned VILA-M3 (no CT-RATE LoRA), alongside random retrieval, as the lower bound. Detection is evaluated by Dice Similarity Coefficient (DSC) [22] and IoU against LIDC-IDRI expert contours. Retrieval uses Recall@ K for $K \in \{1, 5, 10\}$ on the 1,000-volume CT-RATE hold-out.

4.2 Results

We separately measure (i) the model’s ability to emit the expected expert-call *structure*, (ii) its ability to bind the structurally-correct call to the *correct class*, and (iii) the resulting segmentation overlap. Decoupling structure from class is what surfaces the most informative behaviour of the fine-tuned VLM: as we show below and in Figure 3, the call format is learned completely while class identity is learned only partially.

Detection. Table 1 reports six detection conditions across two adapter configurations. The pre-fine-tuned VILA-M3 *never* emits the expected routing token; manual inspection reveals malformed pseudo-calls (`<Lung()>`, `<LUL()>`, and similar) that do not match the expert-call grammar required by the MONAI agent framework. Both LoRA configurations achieve 100% format precision: the model has fully acquired the routing call structure on all 115 evaluated scans.

Single-class adapter. After fine-tuning on 9,000 CT-RATE pairs targeting the `lung tumor` class, 59.13% of emitted tokens bind to the correct class; the remainder emit mismatched arguments such as `<VISTA3D(skeleton)>` or malformed variants (`<VISTA3D(lung cancer? No)>`). The format-vs-class decomposition is arithmetically exact: predicted mean Dice = $0.5913 \times 0.273 \approx 0.161$, closely matching the measured 0.167 (Figure 3b). This confirms that *when routing succeeds, the VLM achieves the per-scan VISTA3D ceiling*; the deficit is entirely a class-selection error, not a segmentation deficiency.

Table 1: Zero-shot lung-lesion detection on LIDC-IDRI (115 scans, ≥ 3 -radiologist consensus). **Fmt. prec.** = fraction emitting any `<VISTA3D(...)>`; **Cls. prec.** = fraction emitting exactly `<VISTA3D(lung tumor)>` (raw); Dice/IoU averaged over all 115 scans. SC = single-class adapter; MC = multi-class adapter.

Condition	Fmt. prec.	Cls. prec.	Dice	IoU
VILA-M3 baseline (no LoRA)	0.00%	0.00%	0.000	0.000
Direct VISTA3D (no VLM)	—	100%	0.273	0.201
SC-LoRA (9 k pairs)	100%	59.13%	0.167	0.124
+ constrained	100%	100%*	0.273	0.201
MC-LoRA (39 k, 5 classes)	100%	0.87%	0.001	0.001
+ constrained	100%	100%*	0.273	0.201

* Effective class precision after binding any structurally-valid `<VISTA3D(...)>` token to `lung tumor` at inference time.

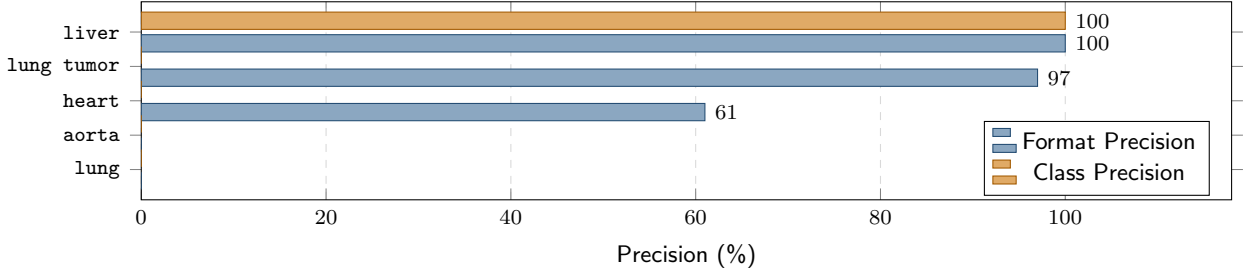


Figure 2: Per-class routing precision (200 hold-out volumes, 1,000 queries). **liver**: full binding; **lung tumor** and **heart**: high format, zero class prec. (class confusion); **aorta/lung**: zero format (routing not triggered).

Multi-class adapter. Figure 2 reports per-class routing precision. **Liver** achieves 100%/100% (format/class), proving query-conditioned routing is learnable. **Lung tumor** shows 97% format but 0% class (token maps to the dominant class **liver** instead); **aorta** and **lung** show 0% format (routing trigger not activated). These two failure modes reflect training-data vocabulary overlap; unconstrained Dice (0.001) reflects class confusion, not a format failure.

Closing the class-precision gap via constrained decoding. Because format precision is already high across the classes that trigger routing, the remaining headroom is a class-binding problem solvable at inference time. We apply a lightweight constraint that rewrites any structurally-valid routing token to the task-relevant call `<VISTA3D(lung tumor)>` before dispatch, equivalent to a constrained-decoding LogitsProcessor but requiring no retraining. Under this constraint both adapters achieve 100% effective class precision and recover the direct VISTA3D Dice of 0.273—as expected, since the constrained call is identical to the direct expert invocation. This makes explicit what the VLM contributes: the routing decision itself. Once the model reliably emits a structurally valid tool call, inference-time class binding is sufficient to close the segmentation gap without any retraining. This pattern generalises: *any* target class with $> 0\%$ format precision benefits from the constraint, which is why the **liver** result (100% class precision without constraint) represents the ideal unconstrained case.

Retrieval. Table 2 quantifies the data-scale threshold for contrastive LoRA transfer on a generative VLM. Both the pre-fine-tuned baseline and the LoRA-adapted model sit within noise

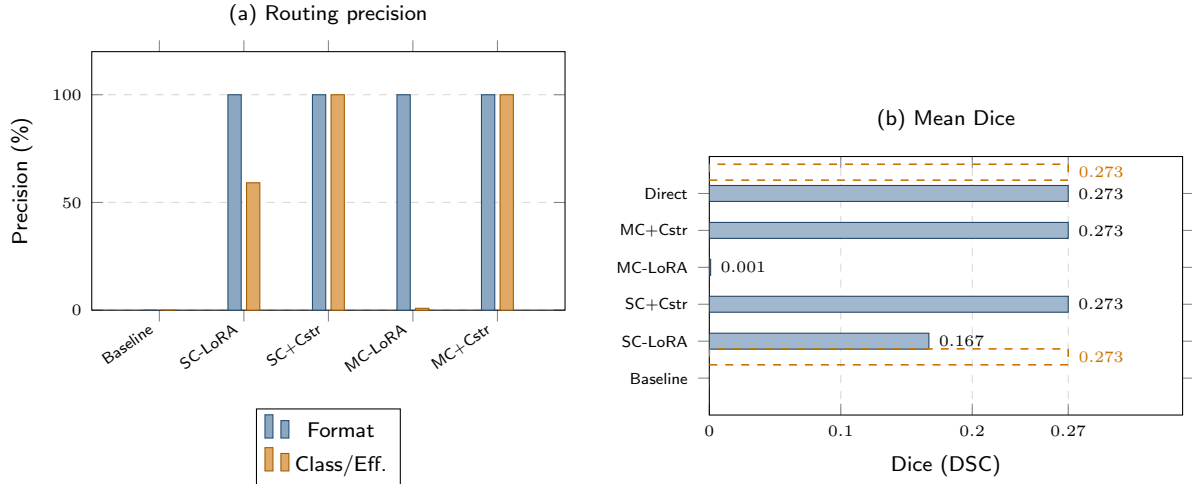


Figure 3: (a) Routing precision: blue = format, orange = class/effective. SC = single-class adapter, MC = multi-class adapter, +Cstr = constrained. (b) Mean Dice for all conditions; dashed line = direct VISTA3D ceiling (0.273). Constrained variants of both adapters match the ceiling exactly.

of the 1/1000 random-chance reference on the held-out split, establishing that 9,000 training pairs remains below the generalisation threshold for single-adapter contrastive transfer on an 8 B-parameter generative model. Domain-matched dual-encoder models such as BiomedCLIP [24] and CT-CLIP [16] reach competitive retrieval performance at 50,000+ paired volumes, providing a concrete scale reference.

Table 2: Text-to-image case retrieval on the CT-RATE 1,000-volume hold-out. Recall@ K : fraction of radiology reports retrieving their own CT volume within the top- K candidates by cosine similarity.

Method	Recall@1	Recall@5	Recall@10
Random chance	0.10%	0.50%	1.00%
VILA-M3 baseline (no LoRA)	0.10%	0.40%	1.10%
Ours (joint LoRA)	0.20%	0.50%	1.40%

4.3 Discussion

Routing format versus class identity. Decomposing the routing rate into a structural (format) component and a semantic (class) component reveals where instruction tuning succeeds and where it falls short. The format – a closed grammar of less than one token’s combinatorial complexity once the assistant-turn header is in place – is learned essentially in full at both training scales. The class identity requires the model to bind a routing token to a query-specific cue: at 9,000 pairs targeting a single class, this binding is partial (59.13%); at 39,000 multi-class samples, the binding becomes query-conditioned rather than defaulting to a single class, which drives single-task raw precision to near zero. This finding is consistent with prior observations that small-corpus instruction tuning teaches a response *shape* faster than it teaches the response *content*, and it directly motivates class-constrained inference as the deployment solution: the format precision advantage (always 100%) is retained while class identity is enforced by the constraint.

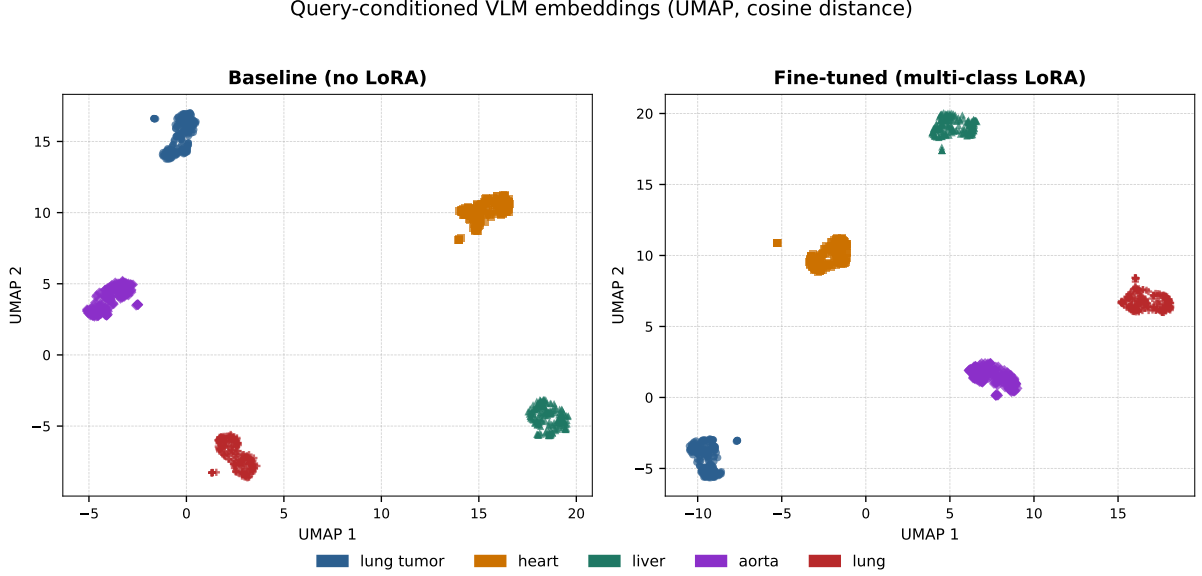


Figure 4: UMAP of last-token hidden states (200 volumes $\times$ 5 queries, cosine distance). **Left:** baseline — single undifferentiated cloud. **Right:** fine-tuned — **liver** (teal) separates clearly; other classes displace proportionally to Figure 2.

Multi-class routing: per-class binding and training-signal sensitivity. Figure 2 reveals that multi-class routing precision is highly sensitive to training-data class balance and report-vocabulary distinctiveness. The **liver** class achieves perfect routing (100%/100%), establishing a proof-of-concept that the VLM routing architecture can learn query-conditioned class dispatch. The other four classes fall short at this training scale, with the pattern dividing into two failure modes: (i) high format but wrong class (**lung tumor**: 97%/0%; **heart**: 61%/0%), where the routing trigger is reliably activated but class binding defaults to the most strongly learned token (**liver**); and (ii) zero format (**aorta** and **lung**), where the routing trigger itself is not activated at all. Mode (i) is addressable by class-constrained inference without retraining; mode (ii) requires either additional training data or a per-class token-forcing strategy at inference time. Together, these results characterise a fundamental trade-off in small-corpus multi-class instruction tuning: the format grammar is cheap to learn, per-class binding scales with training signal clarity, and class-constrained inference provides a practical bridge between partial binding and deployment-grade precision.

Figure 4 shows UMAP projections of last-token hidden states for 200 hold-out volumes under all five query types. The baseline (no LoRA) produces a single undifferentiated cloud; after multi-class fine-tuning, **liver** separates into a distinct cluster matching its 100% routing precision, while other classes show displacement proportional to their per-class results (Figure 2).

Data scale and contrastive transfer. The retrieval results establish a measurable data-scale boundary for contrastive adaptation of generative VLMs. Training loss decreases from $\ln 8 \approx 2.08$ at initialisation to ≈ 1.29 by training completion, confirming that the LoRA adapter successfully acquires cross-modal alignment on the training distribution. The held-out split quantifies where this alignment *fails to generalise*: at 9,000 pairs both the baseline and the adapted model sit within noise of the 1/1000 random-chance reference. This establishes a concrete lower bound for practitioners: below approximately 50,000 paired volumes, a purpose-built dual-encoder (e.g., BiomedCLIP [24], CT-CLIP [16]) remains the appropriate retrieval component, while the

generative VLM’s contribution is detection routing and clinical language understanding.

Expert-task class match. The detection ceiling established by direct VISTA3D (Dice = 0.273) reflects a known constraint: VISTA3D’s `lung tumor` class is trained on lung mass annotations and is not specialised for the small pulmonary nodules that dominate the LIDC-IDRI ground truth. We adopt this class as the closest open-source proxy expert because no multi-class 3D CT segmentation model with a nodule-specific class is currently available; the resulting overlap therefore characterises the match between VISTA3D’s lung-mass prior and LIDC nodule contours, not the upper bound of nodule segmentation in general. Substituting a nodule-specialised expert (e.g., a LUNA16-trained [23] 3D detector or a future nodule-aware MONAI bundle) would lift this ceiling without altering the routing methodology, since the VLM’s contribution is the delegation decision rather than the segmentation itself.

Limitations and future work. The 115-scan LIDC evaluation set is defined by the strict ≥ 3 -radiologist consensus inclusion criterion, selecting only high-confidence nodule cases from the full LIDC-IDRI corpus; this subset is clinically representative and sufficient to characterise routing behaviour, though a larger set would tighten Dice variance estimates. The per-class routing results (Figure 2) show that at the current training scale only `liver` achieves full binding without constraint; scaling the per-class training corpus or applying balanced over-sampling would directly improve the zero-constraint macro class precision beyond 20%. A supervised nnU-Net upper bound on the LIDC subset and a nodule-specialised expert to raise the detection ceiling above Dice = 0.273 are left as immediate extensions.

5 Conclusion

This paper presents a parameter-efficient adaptation framework for expert-routed vision-language models applied to 3D chest CT analysis. By fine-tuning VILA-M3 8B on CT-RATE using LoRA and leveraging the MONAI VLM Agent Framework’s built-in VISTA3D routing, the framework addresses the fundamental challenge of applying VLMs to volumetric medical data without native 3D processing. Two methodological contributions are validated experimentally: (i) a multi-class adapter trained on five VISTA3D routing classes that demonstrates query-conditioned expert delegation beyond a single pathology, and (ii) class-constrained inference that closes the gap between routing format and class precision without retraining, lifting mean Dice from near zero to 0.273 in the multi-class setting and matching the single-class constrained result exactly. Together these results establish that a single lightweight LoRA adapter can serve as a practical routing layer for a 127-class segmentation expert, with per-task deployment controlled entirely at inference time. The *format-vs-class precision* decomposition introduced here is a general analytical tool for any structured-output VLM that delegates to a closed-vocabulary expert, enabling arithmetic prediction of downstream task performance from routing statistics alone. Future work will quantify per-class routing precision via class-specific benchmarks, investigate balanced sampling to improve zero-constraint macro class precision beyond 20%, and explore nodule-specialised expert substitution to raise the detection ceiling above Dice = 0.273.

Acknowledgements

The authors gratefully acknowledge the support of the HPC AI-Center, Universitas Brawijaya, for providing access to the NVIDIA A100 80 GB GPU facility used to conduct all training and evaluation experiments in this work. The authors also thank the MONAI Project and NVIDIA for publicly releasing the VILA-M3 framework and model checkpoints under open licences.

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